USER INTERFACE DESIGN

EXPERIMENT 4: Create a prototype with familiar and unfamiliar navigation elements. Evaluate ease of use with different user groups using proto.io

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AIM:

The aim of this experiment is to develop a Digital Library App prototype incorporating both familiar and novel navigation elements and to assess usability among diverse user groups using Proto.io.

PROCEDURE:

Step 1: Sign Up and Log In

1. Go to https://proto.io
2. Sign up for a new account or log in if you already have one.

Step 2: Create a New Project

1. Click on “Create New Project.”

2. Give the project a name (e.g., “Digital Library App”).
3. Select the device type (Mobile – iPhone X).
4. Click “Create” to start the project.

Step 3: Design the Login Screen

1. Add a New Screen:
    - Click on the “+” button in the left panel.
    - Choose “Blank” and name it “Login”.
2. Add Elements to the Login Screen:
    - Drag a Rectangle to create a header section.
    - Add Text widget and type the app name “LibraByte”.
    - Drag Text Input widgets for:
        - Email/Username
        - Password
            - Drag a Button widget and rename it “Log In”.
            - Add an Image widget to insert a book illustration.
3. Add Interaction:
    - Select the “Log In” button.

- Click on the Interactions tab (⚡ icon).
- Click “+ Add Interaction.”
- Set Trigger as “Tap”.
- Set Action as “Navigate to Screen”.
- Create a new screen named “Home”.

Step 4: Design the Home Screen

1. Add Elements:
    - Change background color to dark blue.
    - Add profile image using Oval shape.
    - Add greeting text (e.g., “Hello User”).
    - Add Search Bar using Rectangle with rounded corners.
    - Add book cover images in horizontal layout.
    - Add genre icons (Fiction, History, etc.).
    - Add Bottom Navigation Bar (Home, Borrowed, Profile).
2. Add Familiar Navigation:
    - Bottom Navigation Bar.
    - Visible Buttons.

    - Search Bar.

3. Add Unfamiliar Navigation:
    - Add Swipe gesture to open hidden menu.
    - Add Long Press interaction on book image to show popup.

Step 5: Link Screens

1. Select a book image on Home screen.
2. Add interaction:
    - Trigger: Tap
    - Action: Navigate to new screen
    - Name the screen “Book Details”.

Step 6: Design Book Details Screen

1. Add:
    - Back Button
    - Book Cover Image
    - Book Title
    - Author Name
    - Description

- “Borrow Now” button

2. Add Interaction:
    - Back Button → Navigate to Home.
    - Long Press on book → Show popup message.

Step 7: Preview the Prototype

1. Click on “Preview” button at top-right corner.
2. Test:
    - Login navigation
    - Home screen navigation
    - Swipe gesture
    - Long press interaction

Step 8: Conduct Usability Testing

1. Define User Groups:
    - Group A – 18–22 years (High tech knowledge)
    - Group B – 30–40 years (Moderate tech knowledge)
    - Group C – 45+ years (Low tech knowledge)

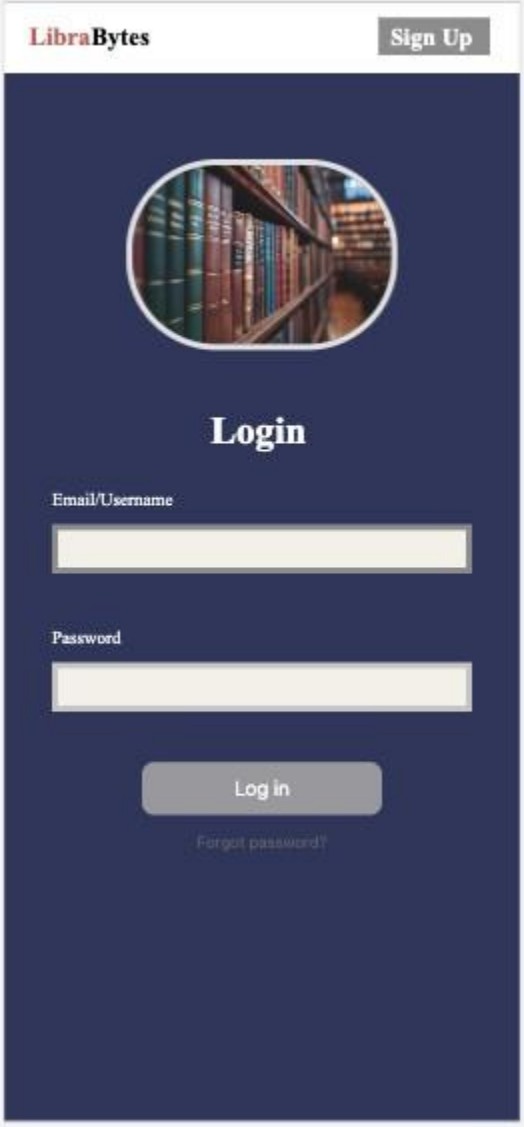
2. Assign Tasks:
    - Log in to the app.
    - Search for a book.
    - Open book details.
    - Use swipe gesture.
    - Navigate back to Home.
3. Collect Feedback:
    - Measure task completion time.
    - Record errors made.
    - Collect satisfaction rating (1–5).

Step 9: Analyze and Evaluate

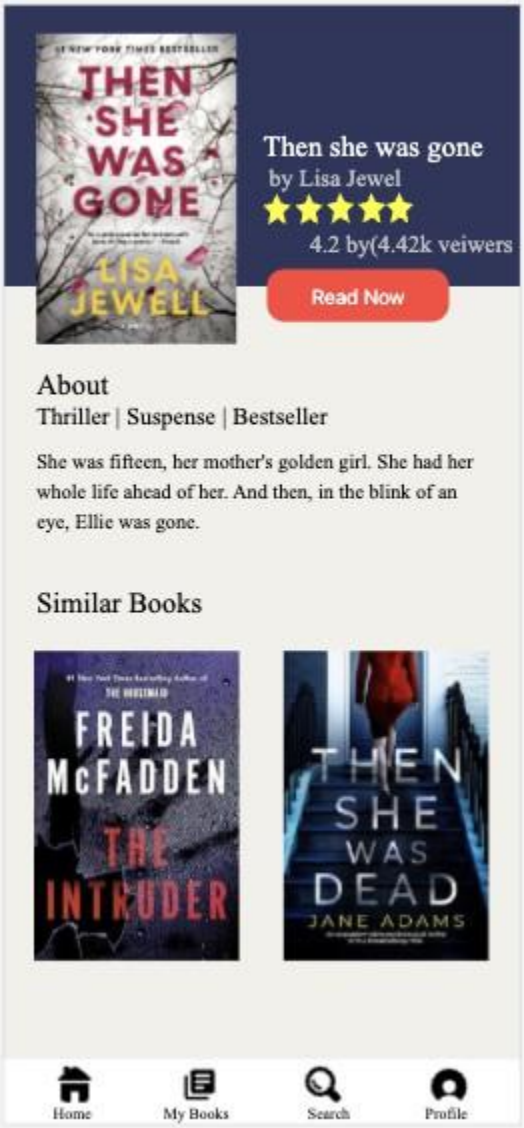
1. Compare how different user groups interacted with familiar and unfamiliar navigation elements.
2. Identify difficulties faced by users.
3. Summarize findings and suggest improvements.

Output:

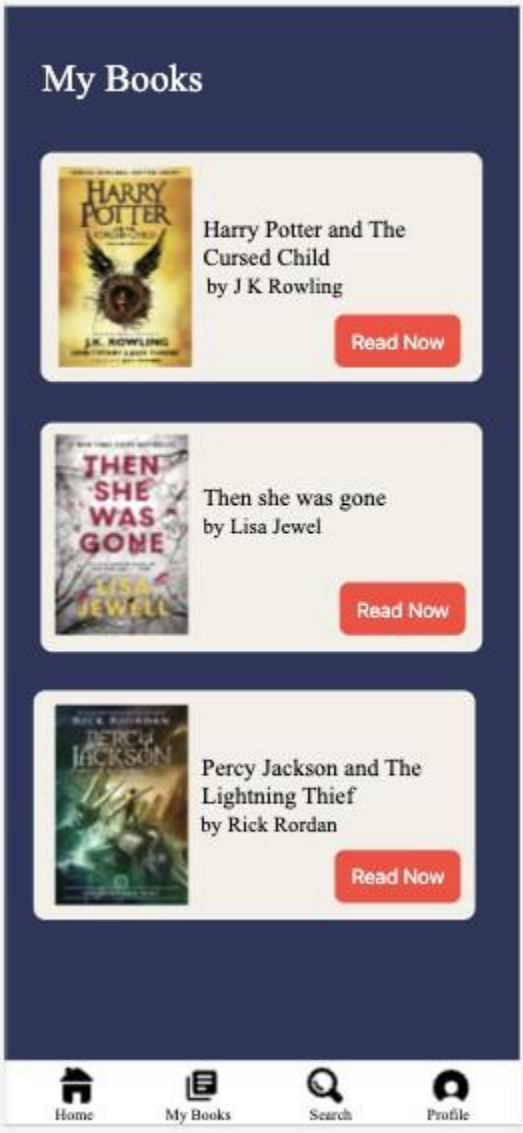
login screen: State 1



home screen: State 1



My books: State 1



Result:

The Digital Library App prototype was successfully developed using Proto.io by incorporating both familiar and unfamiliar navigation elements.